energywave.ai • Humanoid power-budget specification — the load the power supply must meet
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CONFIDENTIAL — DRAFT — FOR DISCUSSION · EVERY SUBSYSTEM LOAD IS AN INPUT, NOT AN ASSERTION

The load, computed, not asserted. Three-month objective #2, as a worksheet.

The deck's claim is specific: a humanoid that runs human-grade sight and sound, smell and taste, touch and motion at once draws more than a lithium battery delivers. This specification turns that claim into numbers that can be checked. It defines the method, lists the published benchmarks with their sources, and leaves every subsystem load as an input to be measured or taken from a datasheet. No figure here is an EnergyWave measurement.

1 · METHOD

Quantity	Definition
Average electrical load $\bar{P}$	Σ over subsystems of (average power while active × duty cycle), plus housekeeping (compute, communications, thermal management, conversion losses), each counted once at a stated measurement boundary.
Peak load P_{peak}	The coincident maximum — actuators under load with sensing and compute active. Sets the supply's power ceiling and ramp rate.
Energy per shift E	$\bar{P}$ × shift length (default 8 h). The number that sizes a pack or a source.
Carried-mass equivalent m	E ÷ pack specific energy (Wh/kg). Compared with what platforms carry today and with the biological benchmark.
Gap	The ratio of E to the energy a platform-sized pack holds at today's pack specific energy; and the runtime a platform-sized pack would give at $\bar{P}$.

2 · BENCHMARKS — EVERY FIGURE ATTRIBUTED, NONE MEASURED BY ENERGYWAVE

Benchmark	Value	Source (status)
Human basal metabolic rate	63–84 W (55-kg woman to 70-kg man)	FAO/WHO/UNU 2004 (reference data)
Human daily expenditure	2.3–4.6 kWh/day (sedentary to vigorous)	FAO/WHO/UNU 2004 (reference data)
Humanoid power at rest	150–400 W; 5–10× less efficient than a human in locomotion	Kashiri et al., Front. Robot. AI 2018 (peer-reviewed review)
Figure 03	2.3 kWh pack; 5 h at peak performance → ≈ 460 W average	Figure AI, "F.03 Battery Development", Jul 2025 (manufacturer claim)
Boston Dynamics Atlas (electric)	4 h; 2 h with heavy lifting; 1.5 h charge	Atlas spec sheet, Dec 2025 (manufacturer claim)
Agility Digit	Up to 4 h; deployed cycles ≈ 90 min run / 9 min charge	Agility, Mar 2025; IEEE Spectrum, Sep 2025 (manufacturer claim; trade press)

Benchmark	Value	Source (status)
Unitree G1 · H1	≈ 0.43 kWh, ≈ 2 h · 0.864 kWh	unitree.com; reseller listing (manufacturer claim)
Tesla Optimus	2.3 kWh, 52 V; “about a full day of work”	Tesla AI Day via IEEE Spectrum, Sep 2022 (manufacturer claim, dated)
Lithium-ion cell specific energy	≈ 220 Wh/kg commercial (2020); 250–271 Wh/kg best cylindrical cells	U.S. DOE Battery500 update 2020; BatteryDesign.net benchmarks (reference data)
Pack specific energy	120–200 Wh/kg; BEV packs ≈ 160–175 Wh/kg in 2026 benchmarks	BatteryDesign.net (reference data)
Body fat	37 MJ/kg ≈ 10.3 kWh/kg; adipose tissue ≈ 8.9 kWh/kg	FAO Food and Nutrition Paper 77; Wishnofsky (reference data)
Battery improvement rate	≈ 4 % per year industry planning assumption	cited in the August 2026 white paper from a patent filing (secondary)

3 · THE WORKSHEET — SUBSYSTEM LOADS TO BE SUPPLIED

Subsystem (deck slide 4)	Components	Average load W	Duty cycle	Basis to be recorded
Sight & sound	Cameras and microphones (sensor draw only; vision and audio processing is counted once, under compute)	**[Needs Input from Ian or Chip]**	**[Needs Input]**	Datasheet or measured draw of the chosen sensor set and the processing it requires.
Smell & taste	Gas-sensor and electrochemical arrays — electronic nose and tongue, including heaters where the sensor needs them	**[Needs Input from Ian or Chip]**	**[Needs Input]**	Datasheet; metal-oxide gas sensors are heater-dominated and must be counted at their heater draw.
Touch	Tactile arrays and electronic skin at nerve-level granularity; readout electronics	**[Needs Input from Ian or Chip]**	**[Needs Input]**	Energy per scan (J) × scans per second × taxel count, or mean power per taxel (W) × count with the duty cycle applied once; readout multiplexing counted in the readout electronics, not per taxel.
Motion	Dense actuators at muscle-level granularity — body and hand; holding a position under gravity draws power unless the actuator or transmission cancels or locks the load (design-dependent)	**[Needs Input from Ian or Chip]**	**[Needs Input]**	Actuator count × average electrical-input power by task profile, measured at the drive input so that motor and drive losses are included once; a potentially major, task-dependent load.
Compute and communications	Perception, planning, control, radios	**[Needs Input from Ian or Chip]**	≈ 1	Datasheet of the chosen compute module at the intended utilization.

Subsystem (deck slide 4)	Components	Average load W	Duty cycle	Basis to be recorded
Thermal management and conversion losses	Fans, pumps, DC–DC conversion	**[Needs Input from Ian or Chip]**	≈ 1	Only losses outside the measured-input boundaries above: DC–DC conversion and thermal management. Kashiri et al. 2018 report ≈ 80 % for a motor example but ≈ 40 % with a typical gearbox and 40–50 % actuator efficiency (their Table 1) — use measured or datasheet efficiencies for the selected architecture, with the electrical / mechanical boundary stated.
Total average load $\bar{P}$	Σ of the rows above. **Anchor for sanity:** the worked manufacturer benchmark — Figure 03's stated 2.3 kWh and 5 h (17 July 2025) — implies ≈ 460 W average, a calculation on a manufacturer's claim; humanoids drew 150–400 W at rest in the 2017-era systems compared by Kashiri et al. 2018 (a dated example, not a contemporary specification).			

4 · WORKED STRUCTURE — WITH THE ANCHOR FIGURE, NOT A COMPANY NUMBER

Step	Using the Figure 03 anchor (≈ 460 W)	With EnergyWave's inputs
Energy per 8-h shift	≈ 3.7 kWh	$\bar{P}$ × 8 h **[Needs Input: $\bar{P}$]**
Pack mass at 160 Wh/kg (pack level)	≈ 23 kg	E ÷ 160 Wh/kg
Pack mass at 250 Wh/kg (cell level, no housing)	≈ 15 kg	E ÷ 250 Wh/kg
Runtime of a 2.3-kWh pack	≈ 5 h, as the manufacturer states	2.3 kWh ÷ $\bar{P}$
Biological comparison	A person's whole day is 2.3–4.6 kWh from fuel at ≈ 10 kWh/kg	E versus the human day; mass ratio at 10 kWh/kg versus 0.16 kWh/kg

The anchor row shows the method on a published pair of numbers; it is not an EnergyWave finding and the platform's task profile is unknown. The deck's premise is that human-grade sensing and actuation push $\bar{P}$ above today's platforms; the worksheet is how that premise becomes a number with a source.

5 · SPECIFICATION OUTPUTS — WHAT THE POWER SUPPLY MUST MEET

Requirement	Value	Derivation
Continuous electrical output	**[Needs Input from Ian or Chip: $\bar{P}$ from §3]**	Average load including housekeeping.
Peak output and ramp	**[Needs Input from Ian or Chip: P_{peak}, ramp rate]**	Coincident actuator peak; fixed in hardware under the control architecture (white paper §4).
Energy per shift	**[Needs Input: E]**	$\bar{P}$ × shift length **[Needs Input from Ian or Chip: shift length and duty profile]**
Mass and volume budget	**[Needs Input from Ian or Chip: target platform mass and bay volume]**	Set by the platform; today's packs occupy the torso.

Requirement	Value	Derivation
Thermal limit	**[Needs Input from Ian or Chip: allowable heat rejection]**	Conversion losses at the stated efficiency.
Target platform	**[Needs Input from Ian or Chip: which humanoid, or a generic class]**	Every row above depends on it.

6 · WHAT THIS SPECIFICATION IS NOT

It does not say that any EnergyWave device meets any row of §5, and it does not describe one. It states the load. Whether a non-chemical source can meet it is the research question the gates exist to answer, and the fourteen orders of magnitude between the picowatt-scale maximum load power inferred from Moddel et al.'s measurements (one laboratory; independent net-cycle validation not established in the reviewed evidence) and the hundreds of watts in §4 are the honest measure of the distance.

Sources: Oct-13 deck slides 3–4 (S-01); Technical Prospectus v1 §§3.1–3.4 and Appendix A refs 32–36 (M-05); Investor Overview slides 3–4 (M-04); DoD white paper v2.0 §§1.1–1.3 (M-07); Physical Architecture & Baseline Model v1.3 (M-14) for the conversion-loss treatment.